

An Assessment of Climate Change Solutions Based on MACs

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Abstract

Climate change is a massive problem with varying degrees of impact. With this impact comes large amounts of solutions. One metric for comparison is the MAC, which reveals the cost per unit emission saved. A comparison is done between climate change solutions, indicating policy-based solutions' large variabilities and behavioral-based energy efficiency's success with this metric.

Keywords: Climate Change, MAC, Solution Comparison, CO2 Equivalent, Cost Analysis

1. INTRODUCTION

Climate change is a massive global threat. It has widespread influence on seemingly unrelated topics, which varies dramatically from rising sea levels and increased frequency of extreme weather events to biodiversity loss and dangers to food supply and security^[1]. It has a massive impact on a wide range of entities - from the present to the future, and from humans to bacteria: all are the impact of human activity. Climate change is caused by the greenhouse effect. This occurs due an increase in atmospheric concentrations of greenhouse gasses, primarily carbon dioxide (CO₂), which trap heat in the Earth's atmosphere and cause global temperatures to rise^[2]. This rise is known to be clearly relatable to human activities^[2].

The worst polluters include the energy, transportation, and agriculture sector^[3]. The energy sector dominates the emissions produced. Within this, power generation and heating accounts for the largest share^[4]. The transportation sector is another significant contributor to emissions. In this category, road transportation takes the blame^[5]. The agriculture sector, despite being less prominent than the previous two sectors, still produces emissions from activities such as deforestation, livestock production, and the use of synthetic fertilizer^[6].

There are many different types of greenhouse gasses causing climate change. As they have different abilities as greenhouse gasses, measuring the sheer amount is an inaccurate way to weigh the impacts being made. To quantify the impact, the carbon dioxide equivalent unit is created. Carbon dioxide equivalent (CO₂e) is a terminology created to compare different greenhouse gasses and their relative impact - considering both quantity and impact potential. As an example, methane (CH₄) is 25 times more potent than CO₂ over a 100-year period. As a

result, 1 kg of methane has the same carbon dioxide equivalent as 25 kg of CO₂. This provides a standard unit for assessing the impacts of greenhouse gasses. In this research, the efficacy of each method will be analyzed in terms of CO₂ equivalents saved. ^{[13] [2]}

CO₂e captures the volume of emissions, thus another factor is used to assess the cost. The cost is based on the MAC (i.e. Marginal Abatement Cost) - a measure of the cost to avoid a certain unit of pollution. This is built upon CO₂e - the unit of pollution in this research is the CO₂e.

Arguably, vast amounts of climate change solutions are present, and an overwhelming amount of research is published regarding these solutions. However, governments and policymakers may lack resources to evaluate these solutions, and due to the lack of research combining these into a practical guide, governments and policymakers would need to either perform tedious analysis (taking time and/or costs) or do less analysis - which would reduce potential climate benefits

As a solution, this research aims to use the CO₂e and MAC to provide valuable insight on the best solutions to implement. This research provides general information in Asian countries, thus further research may be required for certain applications

2. THEORETICAL REVIEW

Climate change is defined as the change in the climate. The definition of 'climate' in its own is vaguely a statistical measure of the weather in a certain area. This definition places a large amount of variations on the specific details, and many models and definitions are known. A set of desiderata for climate change put forward in 2016 (with statements simplified) follows: ^[18]

- A definition must be empirically applicable, thus able to process recorded climates and predict future ones.
- A model must differ between climate conditions uncontroversially regarded as different (e.g. ice age and present climate).
- A model should never rely on knowledge. Enough data is needed to make meaningful predictions, but the definition should be valid without the knowledge.
- The model should apply the same for the past, present, and future.
- A model should be mathematically well-defined (e.g. never reference a non-existent limit)

For the second desideratum, note that the model may classify disputed conditions as either belonging to the same or different type of climate - thus this may be rather dynamic. It will also vary based on the definition of 'uncontroversially regarded'.

Climate change is predicted to increase extreme temperatures more than the GMST (Global Mean Surface Temperature), e.g. warming by 1.5 °C will warm up extreme hot days by 3 °C in some areas. The number of hot days is projected to increase globally, with most additions occurring in the tropics. ^[1]

Meanwhile, sea levels are expected to rise by 10 cm less with a 1.5 °C scenario compared to a 2 °C scenario. However, the rise will continue well past 2100 due to instabilities potentially occurring from warming by 1.5 °C to 2 °C. ^[1]

In terms of land ecosystems, a 1.5 °C scenario will have less negative impacts than 2 °C, and lower the impacts to terrestrial, coastal and freshwater ecosystems - including for conservation and use. ^[1]

Some existing efforts to curb climate change include:

Table 1

Existing Climate Change Mitigation Efforts

No	Effort	Reference
1	Making research results accessible to policymakers	[1] and [3] (e.g. the "Summary for Policymakers"), [2],
2	Publishing research reports to inform decisions and discussions	[4], [5]
3	Improving energy efficiency in buildings and industry	[7]
4	Encouraging sustainable transportation	[8]

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|---|--|---|
| 5 | Encouraging conservation of biodiversity | [9] |
| 6 | Providing support for vulnerable communities and countries | https://www.climatefundsupdate.org/ |
| 7 | Supporting small-scale and community-based efforts | [10] |
| 8 | Providing access to clean energy for all | [11] |
| 9 | Engaging with youth and future generations | [12] |

NOTE. As many references are in APA format as possible, despite some being only a link due to limitations. If the website has been updated, the version(s) used are up on January 2023

One highly critical approach is number 9 - engaging with youth and future generations. People aged 10-24 make up for 1.8 billion of the global population - they are the "largest generation of youth in history". Around 90% live in developing countries, making them a perfect generation for the implementation of climate change solutions. They will play a vital role in protecting the planet, and thus engaging with them is a strategic move. [12]

Another highly important approach is number 7 - supporting small-scale and community-based efforts. Climate change is already taking a toll on the planet, and even if the world becomes carbon-negative at this moment, greenhouse gasses will still take time to reduce. In the meantime, communities and vulnerable areas will need to live with worsened conditions. Thus supporting them is key to their survival.[10]

A third important factor is energy efficiency - including buildings and industries. Energy may come from renewables, but they have limited supply. Additionally, excessive energy use may incentivize use of non-renewables to cope with demand. To prevent this, buildings need to be more efficient. This saves cost, but more importantly eases transition to a greener energy grid. [7]

As visible above, all solutions are unique and work in different ways. By tackling multiple problems arising from this issue it can be more effectively handled.

3. RESEARCH METHOD AND OUTCOME

For this research, secondary data analysis is used. The analysis works by collecting data from existing content published no earlier than 2018. The research uses MAC as the key comparing factor (see Introduction). For the type of data, publications are the source used. This research collects qualitative data regarding the minimum and maximum MAC of the solutions as well as the names given in the original source.

The population being considered is from all publicized climate change solutions i.e. proprietary and closed-access technologies are not considered. However, this is unfeasible for multiple purposes. The first is cost. Climate change solutions have massive variations, and some research on this is paywalled. Accessing all of these would not only be unnecessary, but also expensive. To prevent excessive costs, only free research articles are used. The second is amount. Many free research on climate change exists, and analyzing all would be impossible. As a result, this article picks several sources discerning MACs. An additional constraint on the country being near to Indonesia (the place of this research) is also added. Amongst fulfilling articles, the sample is chosen with no specific precedence.

Data is collected from found publications, accounting for the time collected, and then analyzed. These publications are, in conclusion, free open-access research assessing MACs with the context of neighboring countries.

Table 2

MACs of Climate Change Solutions

Technology / Solution	Min. MAC ** (\$2022 / ton CO ₂ eq)	Max. MAC ** (\$2022 / ton CO ₂ eq)	Reference
Behavioral energy efficiency	-215	-215	[14]
Corn starch ethanol (US)	-20	350	[14]
Renewable Portfolio Standards	0	215	[14]
Reforestation	1	11	[14]
Wind energy subsidies	2	294	[14]
Clean Power Plan	12	12	[14]
Gasoline tax	20	53	[14]
Methane flaring regulation	22	22	[14]
Reducing federal coal leasing	37	76	[14]

CAFE Standards	54	350	[14]
Agricultural emissions policies	56	73	[14]
National Clean Energy Standard	57	124	[14]
Soil management	64	64	[14]
Livestock management policies	80	80	[14]
Concentrating solar power expansion (China & India)	113	113	[14]
Renewable fuel subsidies	113	113	[14]
Low carbon fuel standard	113	3282	[14]
Solar photovoltaics subsidies	158	2376	[14]
Biodiesel	169	282	[14]
Energy efficiency programs (China)	282	339	[14]
Cash for Clunkers	305	475	[14]
Weatherization assistance program	396	396	[14]
Dedicated battery electric vehicle subsidy	396	724	[14]
Solar energy (Thailand, 2015)	59	189	[15]
Wind power (Thailand, 2015)	30	45	[15]
Hydro energy (Thailand, 2015)	-172	-70	[15]
E-fuel (small-scale deployment, projected values until 2050)	1022	1533	[16]
E-fuel (large-scale deployment, projected values until 2050)	25	344	[16]
Low carbon concrete	58	349	[17]

Note. ** MACs are first converted to US Dollars, after which they are accounted for inflation. This avoids bias towards older solutions and ensures fair comparison. All calculations are done to the nearest integer at every step. These values are for reference only, and are used as a rudimentary estimation.

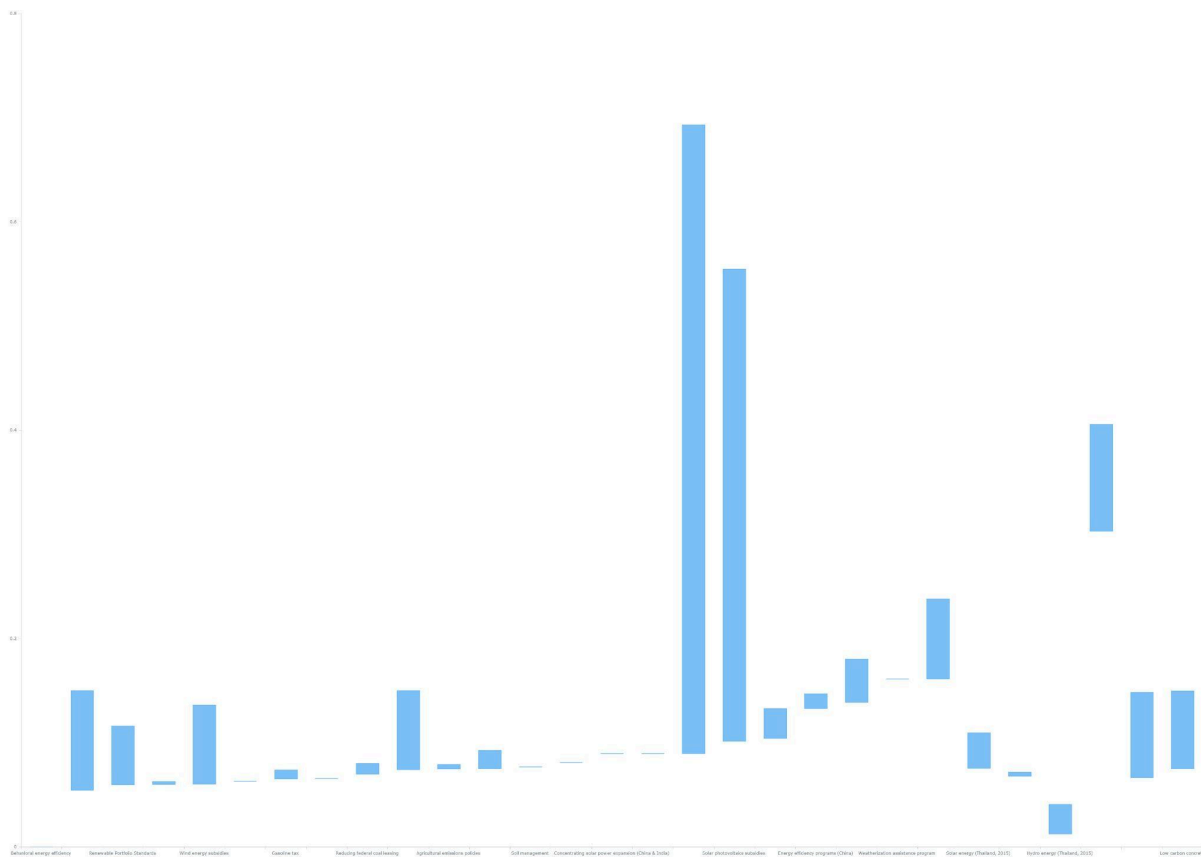
The range of values are noticeably high, making graphical analysis difficult. Plotting the values directly with linear scaling will render most entries unnoticeable. Additionally, some MACs are negative, thus rendering log scales unusable. To fix this issue, normalization is done with the following outline:

- The MACs are linearly projected to a value from 0 to 1, with zero being the minimum and 1 being the maximum. Both minimum and maximum values are considered simultaneously.
- The projections are added by 1 and their natural log is taken as the final normalized value

This eliminates the problem of scaling, and the data can be conveniently plotted. As of the plot itself, the first iteration is a bar chart for the (normalized) minimum and maximum MACs of each solution. However, a span chart is later used as it is more suitable for this purpose.

Due to limitations in the spreadsheet application used to collect the data, plotting as a static image in the document is not possible. Instead, an interactive plot is available at [this link](#) (created and managed by the author). As a confirmation, a preview of the plot is shown:

Preview of final span chart



From these charts, it is clear that "low carbon fuel standards" and "solar photovoltaics subsidies" have the largest variability. Additionally, the lowest MAC belongs to the "behavioral energy efficiency". In the graph it is zero, thus being invisible.

4. DISCUSSION

The two highly volatile solutions are policy-based. These have massive variation based on how the solution is implemented and the conditions of the area. This explains the massive range of MACs.

Apart from this, energy efficiency is a highly affordable solution. This has no cost at all to be done but gives massive reductions.

First, customers and companies will spend less on bills. This reduces the load on the grid and thus allows less infrastructure to be active, used, and maintained. This reduces construction and operation costs. Lastly, this reduces the demand for electricity, allowing less power generation facilities to be used, reducing construction and operation costs linked with having these facilities active. This all provides a net profit/benefit for CO₂ equivalents avoided, thus having a negative MAC. As it has no cost to do, it also has the most benefit, thus having the least MAC (considering the sign). Nevertheless, the MAC may rise if campaigns and educational attempts are done, and this might not always be affordable or even possible.

This research's goal of comparing climate change solutions is successful. A conclusion on the volatility of policy-based solutions, along with the stunning efficiencies of changing people's attitudes are the key outcomes of this research and answers the question. However, deployment scales (e.g. economies at scale) are not yet considered, as the MAC is simply a ratio and does not factor in the deployment size.

This research is a collection of solutions, along with a rudimentary method for their analysis. It is rather versatile and works with any MAC data. Alone this might not be enough to solve the problem but will help interested government entities and policymakers in determining the appropriate solutions for their needs.

Nonetheless, the original plan of collating multiple individual solutions has been achieved, thus this research has met (albeit unlike the ideal plan) the objectives of this research, and may prove helpful for approximate guidance.

5. CONCLUSION

This research answers the goal of comparing climate change solutions using MACs. The research uses free articles and processes the data for plotting. The results show that policy-based solutions have massive variability, while behavioral solutions are likely to be highly efficient. The solutions from multiple resources have been collated as planned.

6. LIMITATION

Although this research may prove helpful for new end-users, there are multiple limits to its direct usefulness. Most readers would have normally accessed articles, thus being able to review the sources directly for better understanding. This research has cut on notes for the sake of simplicity, and leaves some details behind. The research done is reliant on freely accessible (and legal) article copies, which limits the references used. This reduces the direct implications and novelties of the research, and is not of high importance.

Governments and policymakers are the actual target of the solutions, but the sample size used is considerably small due to limited data and preliminary research. This alone would have limited the direct usefulness to them, as some reports have similar amounts of data covered in more detail. Moreover, some data in this article dates back a few years, which means it might have missed the latest developments and thus may be inaccurate. This would further discourage government and policymakers to use the data, with massive public funds at stake.

As an improvement, the first step is to use newer data. With better access to papers this problem might be eradicated. Additionally the data should be processed in a better method, including better accounting calculations, better normalization, and better statistical analysis. Then, better plotting should be done to allow for clearer conclusions and verification. The results can then be published to reputable journals and be used by other researchers.

Additionally, this information may be created into a summary for policymakers (e.g. the "Summary for Policymakers" for IPCC reports). This will attract governments and policymakers in using the data and research with less time with unnecessary technical information, thus lowering the barrier for implementing climate change solutions.

As stated above, one further research is a redo of this research with better methodology and results. Another research possibility is adding other factors such as geographic availability,

maintenance and commissioning costs, implementation scales, etc. This will improve on both theoretical and practical applications of the research.

7. Reference

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